

National Coverage Determination (NCD)

Transcatheter Edge-to-Edge Repair (TEER) for Mitral Valve Regurgitation

20.33

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Tracking Information

Publication Number

100-3

Manual Section Number

20.33

Manual Section Title

Transcatheter Edge-to-Edge Repair (TEER) for Mitral Valve Regurgitation

Version Number

3

Effective Date of this Version

07/31/2023

Implementation Date

07/31/2023

Description Information

Benefit Category

Inpatient Hospital Services

Physicians' Services

Please Note: This may not be an exhaustive list of all applicable Medicare benefit categories for this item or service.

Item/Service Description

A. General

Transcatheter Edge-to-Edge Repair (TEER) of the mitral valve is used in the treatment of mitral regurgitation. TEER approximates the anterior and posterior mitral valve leaflets by grasping them with a clipping device in an approach similar to a treatment developed in cardiac surgery called the Alfieri stitch.

Indications and Limitations of Coverage

B. Nationally Covered Indications

The Centers for Medicare & Medicaid Services (CMS) covers TEER of the mitral valve under Coverage with Evidence Development (CED) with the following conditions:

A. For the treatment of symptomatic moderate-to-severe or severe functional mitral regurgitation (MR) when the patient remains symptomatic despite stable doses of maximally tolerated guideline-directed medical therapy (GDMT) plus cardiac resynchronization therapy, if appropriate, or for the treatment of significant symptomatic degenerative MR when furnished according to a Food and Drug Administration (FDA)-approved indication and when all of the following conditions are met:

1. The procedure is furnished with a mitral valve TEER system that has received FDA premarket approval (PMA).
2. The patient (preoperatively and postoperatively) is under the care of a heart team: a cohesive, multidisciplinary, team of medical professionals. The heart team concept embodies collaboration and dedication across medical specialties to offer optimal patient-centered care. The heart team must include the following members with experience and training as specified:
 - a. Cardiac surgeon
 - i. With ≥ 20 mitral valve surgeries per year or ≥ 40 over two years, 50% of which are mitral valve repairs; and,
 - ii. Who is board eligible or certified in cardiothoracic surgery or similar foreign equivalent.
 - b. Interventional cardiologist
 - i. With professional experience of ≥ 50 career structural heart disease procedures; or ≥ 30 left-sided structural procedures per year; and,
 - ii. With participation in ≥ 20 career trans-septal interventions including 10 as primary or co-primary operator; and,
 - iii. Who is board eligible or certified in interventional cardiology or similar foreign equivalent.
 - c. Interventional echocardiographer (cardiologist or anesthesiologist)
 - i. With professional experience of ≥ 10 trans-septal guidance procedures and ≥ 30 structural heart procedures; and,

- ii. Who is board eligible or certified in transesophageal echocardiography with advanced training as required for privileging by the hospital where the TEER is performed.
 - d. Heart failure cardiologist experienced in treating patients with advanced heart failure (only required for functional MR patients); and,
 - e. Providers from other physician groups as well as advanced patient practitioners, nurses, research personnel, and administrators.
3. Each patient's suitability for surgical mitral valve repair, TEER, or palliative therapy must be evaluated, documented, and made available to other heart team members. Additionally, for patients with functional MR, the heart team heart failure cardiologist must document that the patient has persistent symptoms despite maximally tolerated GDMT and cardiac resynchronization therapy, if appropriate, as described below:
- a. For patients with functional MR: the heart team interventional cardiologist and heart team heart failure cardiologist independently evaluate the patient using information in the medical record and a face-to-face examination. To decrease patient burden, the heart team heart failure cardiologist may meet this requirement through a review of the patient's records and images if the patient has an established relationship with a cardiologist experienced in treating patients with advanced heart failure.
 - b. For patients with degenerative MR: the heart team interventional cardiologist and heart team cardiac surgeon must independently evaluate the patient using information in the medical record and a face-to-face examination.
4. An interventional cardiologist or cardiac surgeon from the heart team must perform the mitral valve TEER and an interventional echocardiographer from the heart team must perform transesophageal echocardiography during the procedure. The interventional echocardiographer may not also furnish anesthesiology during the same procedure. The interventional cardiologist and cardiac surgeon may jointly participate in the intra-operative technical aspects of TEER as appropriate. All physicians who participate in the procedure must have device-specific training as required by the manufacturer.
5. Mitral valve TEERs must be furnished in a hospital with appropriate infrastructure and experience that includes, but is not limited to:
- a. On-site heart valve surgery and interventional cardiology programs;
 - b. Post-procedure intensive care facility with personnel experienced in managing patients who have undergone open-heart procedures;
 - c. Hospital volume requirements below must be met and maintained:
 - i. ≥ 20 mitral valve surgical procedures for severe MR per year or ≥ 40 over two years, of which at least 10 (or 20 over two years) must be mitral valve repairs; and,

- ii. ≥ 2 physicians with cardiac surgery privileges experienced in valvular surgery; and,
- iii. ≥ 1 physician with interventional cardiology privileges; and,
- iv. ≥ 300 percutaneous coronary interventions (PCIs) per year.

6. The heart team and hospital are participating in a prospective, national, audited registry that: 1)comprehensively enrolls TEER patients; 2) accepts all manufactured devices; 3) follows the patient for at least one year; and, 4) complies with relevant regulations relating to protecting human research subjects, including 45 Code of Federal Regulations (CFR) Part 46 and 21 CFR Parts 50 & 56.

The following outcomes must be tracked by the registry, and the registry must be designed to permit identification and analysis of patient, practitioner, and facility level variables that predict each of these outcomes:

- a. Stroke;
- b. All-cause mortality;
- c. Repeat TEER or other mitral procedures;
- d. Transient Ischemic Attacks (TIAs);
- e. Major vascular events;
- f. Renal complications;
- g. Functional capacity; and
- h. Quality of Life (QoL).

7. The registry shall collect all data necessary and have a written executable analysis plan in place to address the following questions. Specifically, for the CED question d, this must be addressed through a composite metric. For the below CED questions (a-e), the results must be reported publicly as described in CED criterion k.

- a. When TEER procedures are performed outside a controlled clinical study, how do outcomes and adverse events compare to the pivotal clinical studies?
- b. How do outcomes and adverse events in subpopulations compare to patients in the pivotal clinical studies?
- c. What is the long-term (≥ 5 year) durability of the device?
- d. What are the long-term (≥ 5 year) outcomes and adverse events?
- e. How do the demographics of registry patients compare to the pivotal studies?

Consistent with section 1142 of the Act, the Agency for Healthcare Research and Quality (AHRQ) supports clinical research studies that CMS determines meet the above-listed standards and address the above-listed research questions.

B. Mitral valve TEERs are covered for uses that are not expressly listed as an FDA-approved indication when performed within a clinical study that fulfills all of the following:

1. An interventional cardiologist or cardiac surgeon must perform the mitral valve TEER and an interventional echocardiographer must perform transesophageal echocardiography during the procedure. The interventional echocardiographer may not also furnish anesthesiology during the same procedure. The interventional cardiologist and cardiac surgeon may jointly participate in the intra-operative technical aspects of TEER as appropriate. All physicians who participate in the procedure must have device specific training as required by the manufacturer.
2. As a fully-described, written part of its protocol, the clinical research trial must critically evaluate the following questions at 12 months or longer follow-up:
 - a. What is the rate of all-cause mortality in the intervention group?
 - b. What is the rate of re-operations (open surgical or transcatheter) of the mitral valve in the intervention group?
 - c. What is the rate of moderate-to-severe or severe MR in the intervention groups?
3. As a fully-described, written part of its protocol, the clinical research study must critically evaluate not only each patient's quality of life pre- and post-TEER (minimum 1 year), but must also address at least one of the following questions:
 - a. What is the incidence of stroke?
 - b. What is the incidence of TIAs?
 - c. What is the incidence of major vascular events?
 - d. What is the incidence of renal complications?
 - e. What is the incidence of worsening MR?
 - f. What is the change in quality of life after TEER?
 - g. What is the change in the patient's functional capacity after TEER?
4. The clinical study must adhere to the following standards of scientific integrity and relevance to the Medicare population:
 - a. The principal purpose of the study is to test whether the item or service meaningfully improves health outcomes of affected beneficiaries who are represented by the enrolled subjects.
 - b. The rationale for the study is well supported by available scientific and medical evidence.
 - c. The study results are not anticipated to unjustifiably duplicate existing knowledge.
 - d. The study design is methodologically appropriate and the anticipated number of enrolled subjects is sufficient to answer the research question(s) being asked in the National Coverage Determination (NCD).
 - e. The study is sponsored by an organization or individual capable of completing it successfully.
 - f. The research study is in compliance with all applicable Federal regulations concerning the protection of human subjects found in the Code of Federal Regulations (CFR) at 45 CFR Part 46. If a study is regulated by the Food and Drug

Administration (FDA), it is also in compliance with 21 CFR Parts 50 and 56. In addition, to further enhance the protection of human subjects in studies conducted under CED, the study must provide and obtain meaningful informed consent from patients regarding the risks associated with the study items and /or services, and the use and eventual disposition of the collected data.

- g. All aspects of the research study are conducted according to appropriate standards of scientific integrity.
- h. The study has a written protocol that clearly demonstrates adherence to the standards listed here as Medicare requirements.
- i. The study is not designed to exclusively test toxicity or disease pathophysiology in healthy individuals. Such studies may meet this requirement only if the disease or condition being studied is life threatening as defined in 21 CFR §312.81(a) and the patient has no other viable treatment options.
- j. The clinical research studies and registries are registered on the www.ClinicalTrials.gov website by the principal sponsor/investigator prior to the enrollment of the first study subject. Registries are also registered in the Agency for Healthcare Quality (AHRQ) Registry of Patient Registries (RoPR).
- k. The research study protocol specifies the method and timing of public release of all prespecified outcomes to be measured including release of outcomes if outcomes are negative or study is terminated early. The results must be made public within 12 months of the study's primary completion date, which is the date the final subject had final data collection for the primary endpoint, even if the trial does not achieve its primary aim. The results must include number started/completed, summary results for primary and secondary outcome measures, statistical analyses, and adverse events. Final results must be reported in a publicly accessible manner; either in a peer-reviewed scientific journal (in print or on-line), in an on-line publicly accessible registry dedicated to the dissemination of clinical trial information such as [ClinicalTrials.gov](https://www.ClinicalTrials.gov), or in journals willing to publish in abbreviated format (e.g., for studies with negative or incomplete results).
- l. The study protocol must explicitly discuss beneficiary subpopulations affected by the item or service under investigation, particularly traditionally underrepresented groups in clinical studies, how the inclusion and exclusion criteria effect enrollment of these populations, and a plan for the retention and reporting of said populations on the trial. If the inclusion and exclusion criteria are expected to have a negative effect on the recruitment or retention of underrepresented populations, the protocol must discuss why these criteria are necessary.
- m. The study protocol explicitly discusses how the results are or are not expected to be generalizable to affected beneficiary subpopulations. Separate discussions in the protocol may be necessary for populations eligible for Medicare due to age, disability or Medicaid eligibility.

Consistent with section 1142 of the Act, the Agency for Healthcare Research and Quality (AHRQ) supports clinical research studies that meet the above-listed standards and address the above-listed research questions.

The principal investigator must submit the complete study protocol, identify the relevant CMS research question(s) that will be addressed, and cite the location of the detailed analysis plan for those questions in the protocol, plus provide a statement addressing how the study satisfies each of the standards of scientific integrity (a. through m. listed above), as well as the investigator's contact information, to the address below. The information will be reviewed, and approved studies will be identified on the CMS Website.

Director, Coverage and Analysis Group
Re: TEER CED
Centers for Medicare & Medicaid Services (CMS)
7500 Security Blvd., Mail Stop S3-02-01
Baltimore, MD 21244-1850

Email address for protocol submissions: clinicalstudynotification@cms.hhs.gov 
Email subject line: "CED TEER [name of sponsor/primary investigator]"

C. Nationally Non-Covered Indications

TEER of the mitral valve is not covered under the following circumstances:

1. For patients in whom existing co-morbidities would preclude the expected benefit from a mitral valve TEER procedure.
2. In patients with untreated severe aortic stenosis.

D. Other

CMS will consider published, peer-reviewed evidence periodically, following the effective date of this NCD and reconsider the policy when appropriate. The NCD will expire 10 years from the effective date if it is not reconsidered during that time. Upon expiration, coverage will be at the discretion of the Medicare Administrative Contractors.

Transmittal Information

Transmittal Number

12112

Coverage Transmittal Link

<https://www.cms.gov/files/document/r12112ncd.pdf> 

Revision History

08/2024 - The purpose of this Change Request (CR) is to provide a quarterly maintenance update of ICD-10 coding conversions and other coding updates specific to National Coverage Determinations (NCDs) and to provide an update to the Appropriate Use Criteria (AUC) Program. No policy is being changed as a result of these updates. ([TN 12757](#)) (CR13706)

06/2022 - Transmittal 10832, dated June 2, 2021, is being rescinded and replaced by Transmittal 11453, dated, June 10, 2022, to revise NCD 90.2, NGS, revises business requirement 12124.2 and 12124.2.1 and its associated spreadsheet of coding by retaining all ICD-10 NOC diagnosis codes proposed for deletion effective July 1, 2022. ([TN 11453](#)) (CR12124)

09/2021 - The purpose of this Change Request (CR) is to inform MACs that on January 19, 2021, CMS expanded coverage of mitral valve TEER procedures for the treatment of functional mitral regurgitation (MR), and maintained coverage of TEER for the treatment of degenerative MR through coverage with evidence development (CED) and with mandatory registry participation. ([TN 10985](#)) (CR12361)

06/2021 - This Change Request (CR) constitutes a maintenance update of ICD-10 conversions and other coding updates specific to NCDs. These NCD coding changes are the result of newly available codes, coding revisions to NCDs released separately, or coding feedback received. ([TN 10832](#)) (CR12124)

05/2021 - This Change Request (CR) constitutes a maintenance update of ICD-10 conversions and other coding updates specific to NCDs. These NCD coding changes are the result of newly available codes, coding revisions to NCDs released separately, or coding feedback received. ([TN 10817](#)) (CR12279)

05/2021 - This Change Request (CR) constitutes a maintenance update of ICD-10 conversions and other coding updates specific to NCDs. These NCD coding changes are the result of newly available codes, coding revisions to NCDs released separately, or coding feedback received. ([TN 10804](#)) (CR12124)

03/2021 - This Change Request (CR) constitutes a maintenance update of ICD-10 conversions and other coding updates specific to NCDs. These NCD coding changes are the result of newly available codes, coding revisions to NCDs released separately, or coding feedback received. ([TN 10624](#)) (CR12124)

01/2021 - Transmittal 10515, dated December 10, 2020, is being rescinded and replaced by Transmittal 10566, dated, January 14, 2021 to remove FISS Reason Codes (RCs) 59041, 59042, 59209, and 59210 from the spreadsheet for NCD 160.18. All other information remains the same. ([TN 10566](#)) (CR12027)

12/2020 - This Change Request (CR) constitutes a maintenance update of ICD-10 conversions and other coding updates specific to NCDs. These NCD coding changes are the result of newly available codes, coding revisions to NCDs released separately, or coding feedback received.

Previous NCD coding changes appear in ICD-10 quarterly updates that can be found at: <https://www.cms.gov/Medicare/Coverage/CoverageGenInfo/ICD10.html>, along with other CRs implementing new policy NCDs. Edits to ICD-10 and other coding updates specific to NCDs will be included in subsequent quarterly releases and individual CRs as appropriate. No policy-related changes are included with the ICD-10 quarterly updates. Any policy-related changes to NCDs continue to be implemented via the current, longstanding NCD process. (TN 10515) (CR12027)

10/2020 - This Change Request (CR) constitutes a maintenance update of ICD-10 conversions and other coding updates specific to NCDs. These NCD coding changes are the result of newly available codes, coding revisions to NCDs released separately, or coding feedback received.

Previous NCD coding changes appear in ICD-10 quarterly updates that can be found at: <https://www.cms.gov/Medicare/Coverage/CoverageGenInfo/ICD10.html>, along with other CRs implementing new policy NCDs. Edits to ICD-10 and other coding updates specific to NCDs will be included in subsequent quarterly releases and individual CRs as appropriate. No policy-related changes are included with the ICD-10 quarterly updates. Any policy-related changes to NCDs continue to be implemented via the current, longstanding NCD process. (TN 10432) (CR12027)

01/2018 - This Change Request (CR) constitutes a maintenance update of International Code of Diseases, Tenth Revision (ICD-10) conversions and other coding updates specific to National Coverage Determinations (NCDs). These NCD coding changes are the result of newly available codes, coding revisions to NCDs released separately, or coding feedback received.

Previous NCD coding changes appear in ICD-10 quarterly updates that can be found at: <https://www.cms.gov/Medicare/Coverage/CoverageGenInfo/ICD10.html>, along with other CRs implementing new policy NCDs. Edits to ICD-10 and other coding updates specific to NCDs will be included in subsequent quarterly releases and individual CRs as appropriate. No policy-related changes are included with the ICD-10 quarterly updates. Any policy-related changes to NCDs continue to be implemented via the current, long-standing NCD process. (TN 2005) (CR10318)

11/2017 - This Change Request (CR) constitutes a maintenance update of International Code of Diseases, Tenth Revision (ICD-10) conversions and other coding updates specific to National Coverage Determinations (NCDs). These NCD coding changes are the result of

newly available codes, coding revisions to NCDs released separately, or coding feedback received. ([TN 1975](#)) (CR10318)

11/2016 - This change request (CR) is the 9th maintenance update of ICD-10 conversions and other coding updates specific to national coverage determinations (NCDs). The majority of the NCDs included are a result of feedback received from previous ICD-10 NCD CR7818, CR8109, CR8197, CR8691, CR9087, CR9252, CR9540, and CR9631. Some are the result of revisions required to other NCD-related CRs released separately.

Edits to ICD-10 and other coding updates specific to NCDs will be included in subsequent, quarterly releases as needed. No policy-related changes are included with these updates. Any policy-related changes to NCDs continue to be implemented via the current, long-standing NCD process. ([TN 1753](#)) (CR9751)

08/2016 - This change request (CR) is the 9th maintenance update of ICD-10 conversions and other coding updates specific to national coverage determinations (NCDs). The majority of the NCDs included are a result of feedback received from previous ICD-10 NCD CR7818, CR8109, CR8197, CR8691, CR9087, CR9252, CR9540, and CR9631. Some are the result of revisions required to other NCD-related CRs released separately.

Edits to ICD-10 and other coding updates specific to NCDs will be included in subsequent, quarterly releases as needed. No policy-related changes are included with these updates. Any policy-related changes to NCDs continue to be implemented via the current, long-standing NCD process. ([TN 1708](#)) (CR9751)

04/2016 - Transmittal 1630, dated February 26, 2016, is being rescinded and replaced by Transmittal 1658 to (1) remove duplicate spreadsheet NCD210.3, (2) add missing spreadsheet NCD20.33, (3) add B/MAC to requirement 3 at request of WPS/B, (4) rename the spreadsheet titles, and, (5) provide a link to the attached spreadsheets for more efficient ease of reference and accessibility. All other information remains the same. ([TN 1658](#)) (CR9540)

12/2014 - The purpose of this Change Request (CR) is to inform contractors payment shall be allowed for Transcatheter Mitral Valve Repair for (TMVR) under Coverage with Evidence Development.

National Coverage Analyses (NCAs)

This NCD has been or is currently being reviewed under the National Coverage Determination process. The following are existing associations with NCAs, from the National Coverage Analyses database.

- [Original Consideration for Transcatheter Mitral Valve Repair \(TMVR\) \(CAG-00438N\)](#) 
- [First reconsideration for Transcatheter Mitral Valve Repair \(TMVR\) \(CAG-00438R\)](#) 

Additional Information

Other Versions

Title	Version	Effective Between	
Transcatheter Edge-to-Edge Repair (TEER) for Mitral Valve Regurgitation	3	07/31/2023 - N/A	You are here
Transcatheter Edge-to-Edge Repair (TEER) for Mitral Valve Regurgitation	2	01/19/2021 - 07/31/2023	View
Transcatheter Mitral Valve Repair (TMVR)	1	08/07/2014 - 01/19/2021	View